Unit 5 – Polynomial Functions – Day 11 *Accelerated II/III*

Irrational & Complex Root Theorems

Irrational Root Theorem: If $a+\sqrt{b}$ is a root (zero) of a polynomial, then the conjugate _____ is also a root.

Complex/Imaginary Root Theorem: If $a+bi$ is a root (zero) of a polynomial, then the conjugate _____ is also a root.

Example 1: Given the following roots, find the other additional roots.

A) $1+\sqrt{3};\ -\sqrt{11}$ B) $2-\sqrt{7};\ 4+5i$ C) $3-i;\ 4i$

Example 2: Find a third-degree polynomial equation with integer coefficients that has the roots *8* and *3i*.

Example 3: Find a polynomial equation with integer coefficients and as the roots $5\ \&\ \sqrt{2}$.

Example 4: Find a polynomial equation with integer coefficients and as the roots -3 & $1+2i$.

Example 5: Find a polynomial equation with integer coefficients and as the roots $\frac{3}{4}$ & $2-\sqrt{5}$.